## EXERCISE 4.5

**Divide the following.**

| | | |
|---|---|---|
| (1) 17.1 ÷ 6 | (2) 1.8 ÷ 5 | (3) 9.36 ÷ 4 |
| (4) 9.66 ÷ 3 | (5) 4.68 ÷ 9 | (6) 17.85 ÷ 7 |
| (7) 6.28 ÷ 4 | (8) 0.72 ÷ 2 | (9) 5.62 ÷ 2 |
| (10) 34.23 ÷ 3 | (11) 67.25 ÷ 5 | (12) 32.34 ÷ 6 |
| (13) 96.20 ÷ 2 | (14) 10.11 ÷ 3 | (15) 40.05 ÷ 5 |

**Solve real life problems involving decimals up to two decimal places.**

**Example 1:** Sana bought **0.25 kg** of cadbury chocolates, **10.50 kg** of flour and **2.50 kg** of sugar. What is the total mass of ingredients she bought?

**Solution:**

| | | |
|---|---|---|
| Cadburry chocolate | $^{1}$0.25 | Kg |
| Flour | 10.50 | Kg |
| Sugar | + 2.50 | Kg |
| | **13.25** | Kg |

She bought a total mass of **13.25 kg**.

**Example 2:** Ali's height is **1.75 m** and Azhar's height is **1.27 m**. How tall is Ali than Azhar?

**Solution:**

| | |
|---|---|
| Ali's height | 1 . $\overset{6}{\cancel{7}}$ $^{1}$5 |
| Azhar's height | − 1 . 2 7 |
| Difference | **0 . 4 8** |

Hence Ali is **0.48 m** taller than Azhar.

### Example 3:

The cost of one kilogram of flour is **Rs 32.50**, what will be the total cost of **15 kg** of flour?

### Solution:

One kg of flour cost = **Rs 32.50**,
the cost of **15 kg** will be = **32.50 x 15**

So,

$$\begin{array}{r} 32.50 \\ \times \quad 15 \\ \hline 16250 \\ +32500 \\ \hline 487.50 \\ \hline \end{array}$$

The total cost of 15 kg of flour will be **Rs 487.50**

### Example 4:

Mrs. Aslam wants to divide **3.36kg** of sweets equally among **4** relatives. How much will each relative get?

### Solution:

$$\begin{array}{r} 0.84 \\ 4\overline{)3.36} \\ -32 \\ \hline 16 \\ -16 \\ \hline 00 \end{array}$$

[ before taking 33, shift the decimal point to the quotient ]

$3.36 \div 4 = 0.84$

Hence, each relative will get **0.84** kg of sweets.

## EXERCISE 4.6

1. Ahmed purchased a shirt for **Rs 325.80** and a jeans for **Rs 525.25.** Find the amount spent.

2. Javeria's weight is **10.24 kg** and her sister's weight is **11.28 kg**. What is the total weight of both the sisters?

3. Ansa paid a total cost of **Rs 97.5** to the shopkeeper for a pencil case and a colour box. If the pencil case costs **Rs 30.25,** find the cost of the colour box.

4. Haroon and Shafique spent **Rs 95.82.** Haroon spent **Rs 89.75.** What amount was pent by Shafique?

5. Ali has Rs 50.29 as pocket money. He gave **Rs 15.45** to his sister Nida and spent **Rs 13.84** on coffee. How much money has left with him?

6. Aslam brought **65** hens for his farm. The weight of each hen is **2.72 kg**. What is the weight all have?

7. One set of books weigh **3.75 kg.** What will be the weight of **32** such sets?

8. Yusra cuts a ribbon **13.75 m** long into **5** equal pieces. Find the length of each piece.

9. The total weight of **5** sacks of flour is **58.75 kg**. Find the weight of one sack of flour.

## REVIEW EXERCISE – 4

1. **Write down the place value and value of circled digits in the following.**

(i) 2.325 (ii) 17.953 (iii) 37.87

2. **Convert the following fractions into decimals.**

(i) $\frac{321}{100}$ (ii) $\frac{175}{1000}$ (iii) $\frac{19}{4}$ (iv) $\frac{27}{8}$

3. **Convert the following decimals into fractions.**

(i) 1.54 (ii) 0.35 (iii) 13.7 (iv) 0.345

4. **Solve the following.**

(i) 1.57 + 3.42 (ii) 0.37 − 0.18

(iii) 17.55 + 23.42 (iv) 1.94 − 0.89

5. **Multiply the following.**

(i) 0.325 x 100 (ii) 17.55 x 10

(iii) 3.5252 x 1000 (iv) 2.57 x 32

6. **Divide the following.**

(i) 9.42 ÷ 6 (ii) 2.52 ÷ 7 (iii) 8.61 ÷ 3

7. Anas ran a distance of **110.25m** and Asman ran **97.75m**.
a. What is the total distance that they both ran altogether?
b. How much more distance did Anas ran than Asma?

8. Abdullah drinks **0.45** litre of juice every day.
How many litres of juice does he drinks in **30** days?

Unit 5

# MEASUREMENTS

## 5.1 LENGTH

We have learnt that the small lengths are measured in metres (m) and centimetres (cm) and long distances are measured in kilometres (km).

### 1. Conversion of units of length:

**Convert kilometres to metres, metres to centimetres and centimetres to millimetres.**

### (i) Convert kilometres to metres.

There are 1000 metres in a kilometre.

**1 kilometre (km) = 1000 metre (m)**

So, we multiply the numbers of kilometres by 1000 to change them into meters.

**Example 1:** Convert 2 kilometres into metres

*Solution:* 2 km = 2 x 1000 = 2000 m

**Example 2:** Convert 8 kilometres into metres

*Solution:* 8 km = 8 x 1000 = 8000 m

**Activity 1** **Convert to meters.**

| | | | | |
|---|---|---|---|---|
| 1 | 6 km = | 6 x 1000 | = 6000 | m |
| 2 | 9 km = | ________________ | = ______ | m |
| 3 | 12 km = | ________________ | = ______ | m |
| 4 | 25 km = | ________________ | = ______ | m |
| 5 | 50 km = | ________________ | = ______ | m |
| 6 | 75 km = | ________________ | = ______ | m |
| 7 | 80 km = | ________________ | = ______ | m |

**Teacher's Note** Teacher should explain the students to convert the different units of length through multiplication by 1000, 100 and 10.

### (ii) Convert metres to centimetres

There are 100 centimetres in a metre.

**1 metre (m) = 100 centimetres (cm)**

So, we multiply metres by 100 to change into centimetres.

**Example 1:** Convert 6 metres into centimetres

*Solution:* 6 x 100 = 600 cm

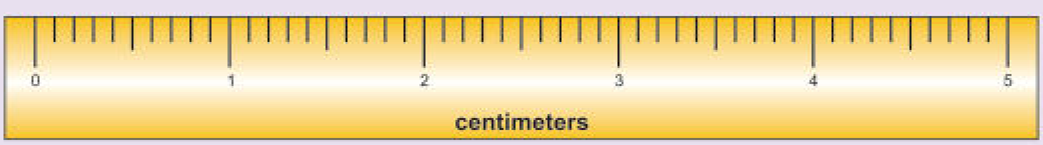
**Example 2:** Convert 20 metres into centimetres

*Solution:* 20 m = 20 x 100 = 2000 cm

**Activity 2** Convert metres into centimetres.

| | |
|---|---|
| 1 | 5 m = ___5 x 100___ = ___500___ cm |
| 2 | 9 m = ______________ = ______ cm |
| 3 | 18 m = ______________ = ______ cm |
| 4 | 45 m = ______________ = ______ cm |
| 5 | 60 m = ______________ = ______ cm |
| 6 | 87 m = ______________ = ______ cm |

### (iii) Convert centimetres to millimetres.



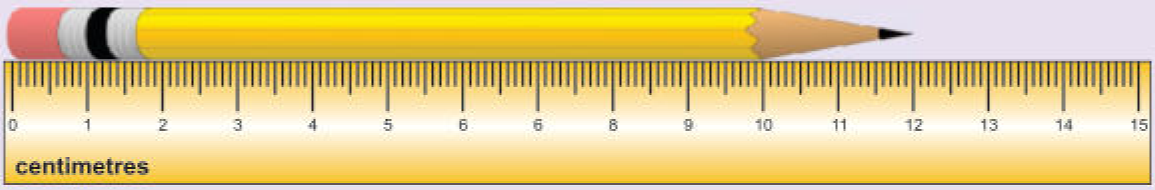

Each centimetre unit is divided into 10 smaller units.

Each smaller unit is called a **millimetre (mm).**

There are 10 mm in a cm.

**1 centimetre (cm) = 10 millimetres (mm)**

**Example 1:** Measure the length of this pencil in centimetres and convert it into millimetres.




*Solution:* The length of this pencil is 12 cm.

To find the length in millimetres, we multiply centimetres by 10.

So, **12 cm = 12 x 10 = 120 mm**

The length of pencil is 120 mm.

**Example 2:** Convert 85 cm to millimetres

*Solution:* 85 cm = 85 x 10 = 850 mm

**Activity** **Convert the following into millimetres.**

| | | | |
|---|---|---|---|
| 1 | 7 cm = 7 x 10 | = 70 | mm |
| 2 | 10 cm = ________ | = ____ | mm |
| 3 | 11 cm = ________ | = ____ | mm |
| 4 | 31 cm = ________ | = ____ | mm |
| 5 | 49 cm = ________ | = ____ | mm |

Following is the table showing relationship among units of length.

**10 millimetres = 1 centimetre**

**100 centimetres = 1 metre**

**1000 metres = 1 kilometres**

**1 metre = 1000 millimetres**

**Example 3:** A boy purchased a rope 2 metre long. Convert it into centimetres and millimetres.

2 metre

*Solution:* 2 m = 2 x 100 cm = 200 cm

*Again:* 2 m = 2 x 1000 mm = 2000 mm

## EXERCISE 5.1

**1 Convert into metres.**

(i) 5 km (ii) 14 km (iii) 20 km

**2 Convert into centimetres.**

(i) 17 m (ii) 32 m (iii) 54 m

**3 Convert into millimetres?**

(i) 15 cm (ii) 19 cm (iii) 30 cm

**4 Change into centimetres and millimetres?**

(i) 4 m (ii) 10 m (iii) 35 m

(iv) 64 m (v) 83 m (vi) 98 m

## 2. Addition and Subtraction of units of length:

### Add and subtract expressions involving similar units of length.

**As metres are added to metres and kilometres are added to kilometres, so like units are to be added and subtracted from each other.**

**Example 1:** Add **24 km 233 m** and **20 km 446 m**

*Solution:*

| | km | m |
|---|---|---|
| | 24 | 233 |
| + | 20 | 446 |
| | 44 | 679 |

Thus, sum is **44 km 679 m**

**Example 2:** Add **42 m 75 cm** and **28 m 90 cm**

*Solution:*

| | m | cm |
|---|---|---|
| | 1 1 | |
| | 42 | 75 |
| + | 28 | 90 |
| | 71 | 65 |

Thus, sum is **71 m 65 cm**

**Example 3:** Subtract **34 km 23 m** from **78 km 86 m**

*Solution:*

| | km | m |
|---|---|---|
| | 78 | 86 |
| – | 34 | 23 |
| | 44 | 63 |

Thus, difference is **44 km 63 m**

**Example 4:** Subtract **25 m 56 cm** from **47 m 23 cm**

*Solution:*

| | m | cm |
|---|---|---|
| | $4\overset{6}{\cancel{7}}$ | $\overset{1}{\cancel{2}}3$ |
| – | 25 | 56 |
| | 21 | 67 |

Thus, difference is **21 m 67 cm**

## EXERCISE 5.2

**(1) Add:**

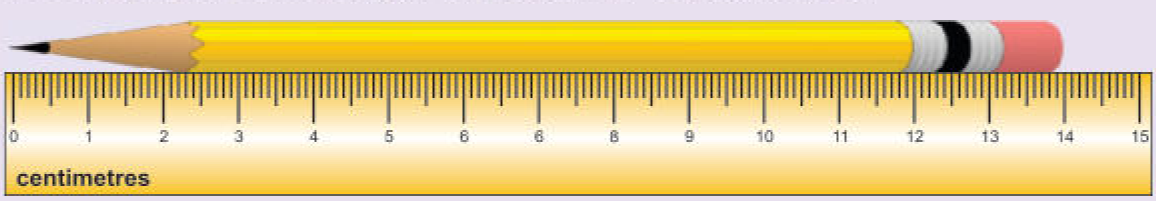
(i) 4200 m and 9600 m
(ii) 25 km 520 m and 12 km 840 m
(iii) 49 km 719 m and 32 km 103 m
(iv) 30 km 60 m and 29 km 29 m
(v) 69 m 17 cm and 99 m 32 cm
(vi) 42 cm 3 mm and 68 cm 5 mm
(vii) 13 m 25 cm, 40 m and 65 m 5 cm
(viii) 90 km 820 m, 75 km 500 m and 110 km 175 m
(ix) 45 km 340 m , 82 km 399 m and 230 km 180 m

**(2) Subtract:**

(i) 5050 m from 7000 m
(ii) 2 m 76 cm from 6 m 35 cm
(iii) 34 m 20 cm from 36 m 80 cm
(iv) 305 m 20 cm from 862 m 60 cm
(v) 36 km 500 m from 87 km 250 m
(vi) 18 km 352 m from 70 km 100 m
(vii) 106 m 18 cm from 300 m 29 cm
(viii) 27 cm 8 mm from 74 cm 7 mm
(ix) 37 cm 5 mm from 64 cm 3 mm

### Use appropriate units to measure the length of different objects

We measure the length of pencil in centimetres.




The length of pencil is 14 cm.

We measure the length of book in centimetres.

We measure the length of table, room or play ground in metres.

We measure the distance between two cities in kilometres.

# Unit 5 MEASUREMENTS (Length)

**Activity** **Use the correct unit of cm , m and km to fill the each box.**

**We measure:**

| | | |
|---|---|---|
| 1 | The length of pen in | cm |
| 2 | The length of bed in | |
| 3 | The length of hockey ground in | |
| 4 | The width of your geometry box in | |
| 5 | The distance from Karachi to Larkana in | |

**Example: Tick (✓) the best unit of length for measuring the length of sides of table**

**(a) 45 mm** **(b) 45 cm ✓**
**(c) 45 m** **(d) 45 km**

## Solve real life problems involving conversion, addition and subtraction of units of length

**Example 1:** Farhan is 1 m 30 cm tall. He stands on a stool 70 cm high. How high is the top of his head from the ground?

*Solution:*

| | m | cm |
|---|---|---|
| Farhan's height | (1) 1 m | 30 cm |
| Height of stool | + | 70 cm |
| | **2 m** | **00 cm** |

[Sum of (30 + 70) cm = 100 cm
100 cm = 1 m
carry 1 m to metre column]

The top of his head is 2 m high from the ground.

**Example 2:** There are 2 pieces of wood. The first piece measures 5 m 28 cm and the second piece is 3 m 55 cm long. What is the difference between the two?

*Solution:*

| | m | cm |
|---|---|---|
| Length of the first piece of the wood | 5 m | 28 cm |
| Length of the second piece of the wood | − 3 m | 55 cm |
| | **1 m** | **73 cm** |

The difference between the two pieces of wood is **1** m **73** cm.

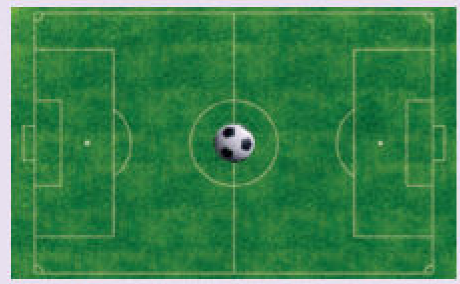
## EXERCISE 5.3

1 Choose the best unit of length for the following objects:

(i) **The thickness of eraser is:**


(a) 2 mm (b) 2 cm
(c) 2 m (d) 2 km

(ii) **The length of football ground:**



(a) 30 mm
(b) 30 cm
(c) 30 m
(d) 30 km

(iii) **The distance between Karachi and Hyderabad by road is:**

(a) 165 mm
(b) 165 cm
(c) 165 m
(d) 165 km


2 The length of an iron rod is 2 m 86 cm. How much iron rod is left if 1m 38 cm has been cut off?

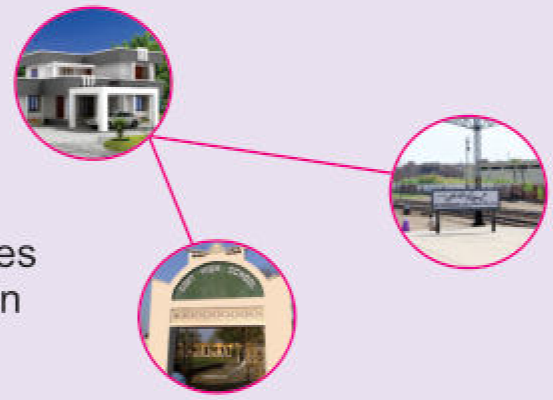
3 Ali covered a distance of 789 m from his house to Jamia Masjid and then 368 m from Jamia Masjid to School. Find the total distance covered by him ?

4 A car is 1m 62 cm wide. A garage is 2 m 41cm wide. How much space is left when the car is in the garage?

5 The red part of a colour pencil is 65 mm long. The blue part is 52 mm long. What is the length of full pencil in millimetres and centimetres?

6 In a walking race, in specified time Tariq ran 9 km 200 m, Sajjad ran only 8 km 850 m. How far ahead of Sajjad was Tariq?

7. In a 10 kilometre race, a horse fall down at a distance of 245 m from the winning point. What distance had the horse run before it fall down?
8. Nasir is 142 cm tall. His friend is 8 cm taller than Nasir. How tall is his friend?
9. Saba's house is at a distance of 375 m from school and 505 m from railway station. What is the difference between distances of school and railway station from Saba's home?



## 5.2 MASS / WEIGHT

**The unit of mass is kilogram (kg). Kilograms (kg) are used to weigh heavy objects and grams are used to weigh light objects.**

5g 10g 50g 100g 200g 500g

**Convert kilograms to grams.**

There are 1000 grams in a kilogram.

**1 kilogram (kg) = 1000 gram (g)**

So, we multiply the numbers of kilograms by 1000 to convert into grams.

**Example 1:** Convert 15 kg to grams

*Solution:* 15 kg = 15 x 1000 g = 15000 g

**Example 2:** Convert 2 kg 250 g into grams

*Solution:* 2 kg 250 g = 2 x 1000 g + 250 g

= 2000 g + 250 g = 2250 g

**Teacher's Note** Teacher should explain students to convert the different units of kilograms to gram by multiplication.

**Activity** Convert the following into grams.

(1) **18 kg** = 18 x 1000 = 18000 g
(2) **25 kg** = ________ = ________ g
(3) **4 kg 80 g** = ________ = ________ g
(4) **5 kg 125 g** = ________ = ________ g

## 3. Addition and subtraction of unit of Mass/Weight

### Add and subtract expressions involving similar units of mass/weight.

**Example 1:**
Add **3 kg 65 g** and **5 kg 30 g**.

*Solution:*

| | kg | g |
|---|---|---|
| | 3 | 65 |
| + | 5 | 30 |
| | **8** | **95** |

Total weight = **8 kg 95 g**

**Example 2:** Subtract **7 kg 650 g** from **9 kg 500 g**.

*Solution:*

| | kg | g |
|---|---|---|
| | 8 | 14 10 |
| | ~~9~~ | ~~5~~00 |
| − | 7 | 650 |
| | **1** | **850** |

Total weight = **1 kg 850 g**

## EXERCISE 5.4

**1 Add.**

(i) 3705 g, 8536 g and 4000 g
(ii) 4 kg 485 g, 2 kg 390 g and 4 kg 425 g
(iii) 8 kg 75 g and 9 kg 46 g
(iv) 4 kg 32 g and 3 kg 85 g
(v) 16 kg 860 g, 23 kg 545 g and 49 kg 360 g

**2 Subtract.**

(i) 1 kg 250 g from 5 kg
(ii) 3 kg 33 g from 6 kg 86 g
(iii) 4505 g from 9007 g
(iv) 36 kg 740 g from 59 kg 960 g
(v) 14 kg 72 g from 20 kg 40 g

## Use appropriate units to measure the mass/weight of different objects.

- To weigh heavy objects we use kilogram.
- To weigh lighter objects we use gram.

**For example:**

We know the vegetable seller is weighing potatoes in kilogram (kg). The goldsmith is weighing a ring in grams (g). The wheat bags are weighing in kilogram (kg).

**Activity** Use appropriate unit in the blanks 'kg' or 'g'.

1. The mass of a silver ring is measured in ____g____
2. The mass of bags of flour are measured in ____________
3. The mass of sugar bag is measured in ____________
4. The mass of one soap is measured in ____________
5. The mass of potatoes and onions are measured in ______
6. The mass of salt bag is measured in ____________

## Solve real life problems involving conversion, addition and subtraction of units of mass/weight.

**Example 1:** A rice merchant sold 168 kg 750 g of rice and had 57 kg 650 g left. Find the quantity of rice in the beginning?

*Solution:*

| | | kg | g |
|---|---|---|---|
| Weight of rice sold | | 168 | 750 |
| Weight of rice left | + | 57 | 650 |
| | | **226** | **400** |

He had 226 kg 400 g rice in the beginning.

**Example 2:** A hen weighs 2 kg 720 g and a duck weighs 4 kg 240 g. How much duck is heavier than the hen?

*Solution:*

| | kg | g |
|---|---|---|
| Weight of the duck | 4 | 240 |
| Weight of the hen | − 2 | 720 |
| | **1** | **520** |

The duck is 1 kg 520 g heavier than the hen.

## EXERCISE 5.5

**1 Choose the answer in best unit of mass/weight for following objects:**

(i) The mass of a paper clip.
**(a) 1 g (b) 1 kg (c) 100 g (d) 100 kg**

(ii) The mass of a 13 years old boy.
**(a) 4 g (b) 4 kg (c) 40 kg (d) 400 g**

(iii) The mass of a box of tea.
**(a) 40 g (b) 400 g (c) 4 g (d) 4 kg**

(iv) The weight of a watermelon.
**(a) 5 kg (b) 50 g (c) 50 kg (d) 500 g**

2 A bale of rubber weighs 75 kg 700 g. Another weighs 86 kg 400 g. Find their total weight.

3 Hussain weighs 28 kg 750 g and his father weighs 63 kg 500 g. How much lighter is Hussain than his father?

4 Fozia bought 21 kg 350 g of sweet from one shop. She purchased 1 kg 200 g of sweet from another shop. Find the total weight of sweet she purchased in all?

5 A grain merchant had 3000 kg of peas. He sold 1856 kg 750 g of it. What weight of peas had he left?

## 5.3 VOLUME / CAPACITY:

### (1) Conversion of units of capacity

The basic unit to measure capacity is litres $(l)$. The smaller unit to measure the capacity is millilitre $(ml)$.

**Convert litres to millilitres.**

There are 1000 millilitres in a litre.

**1 litre $(l)$ = 1000 millilitres $(ml)$**

In order to convert a litre into millilitres we multiply the number of litres by 1000.

**Example 1:** Convert 8 litres into millilitres

*Solution:* 8 l = 8 x 1000 l = 8000 ml

**Activity** **Convert the following into millilitres.**

(1) **15** $l$ = **15 x 1000** = **15000** $ml$.

(2) **40** $l$ = ________ = ________ $ml$.

(3) **75** $l$ = ________ = ________ $ml$.

(4) **66** $l$ = ________ = ________ $ml$.

### (2) Addition and Subtraction of Units of Capacity

**Add and subtract expression involving units of capacity/volume**

**Example 1:** Add 9 $l$ 800 $ml$ and 2 $l$ 300 m$l$.

*Solution:*

$$\begin{array}{rr} l & ml \\ \overset{1}{9} & 800 \\ +\,2 & 300 \\ \hline \mathbf{12} & \mathbf{100} \\ \hline \end{array}$$

Total volume = **12** $l$ **100** $ml$.

**Teacher's Note** Teacher should explain the students to convert litres into millilitres by multiplication.

## EXERCISE 5.6

**(1) Add.**

(i) 7 $l$ 420 $ml$ and 10 $l$ 500 $ml$

(ii) 2100 $ml$, 4960 $ml$ and 3755 $ml$

(iii) 7 $l$ 25 $ml$, 16 $l$ 400 $ml$ and 31 $l$ 251 $ml$

(iv) 705 $ml$, 820 $ml$ and 695 $ml$

(v) 14 $l$ 782 $ml$, 17 $l$ 300 $ml$ and 26 $l$ 450 $ml$

**(2) Subtract.**

(i) 719 $l$ from 825 $l$

(ii) 16 $l$ 415 $ml$ from 60 $l$ 600

(iii) 640 $ml$ from 905 $ml$

(iv) 8 $l$ 205 $ml$ from 11 $l$ 150 $ml$

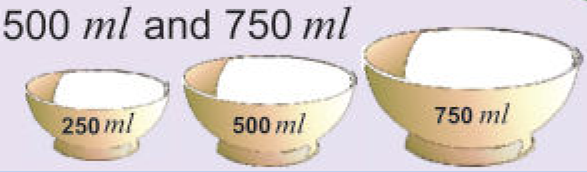
(v) 76 $l$ 223 $ml$ from 97 $l$ 660 $ml$

**Use appropriate units to measure the capacity/volume of different objects (utensils etc).**


- The capacity of milk pack is $\frac{1}{2}$ $l$ or 500 $ml$.
- The capacity of 2nd milk pack is 1 $l$ or 1000 $ml$.
- The capacity of mineral water bottle is 1 litre.



- The capacity of bucket is 5 $l$ or 5000 $ml$.
- The capacity or volume of bowls are 250 $ml$, 500 $ml$ and 750 $ml$



**Activity** **Choose the appropriate measure of given objects.**

1


2


3



| 1 | 2 | 3 |
|---|---|---|
| 25 $ml$ | 36 $ml$ | 1 $l$ |
| 250 $ml$ ✓ | 360 $ml$ | 10 $l$ |
| 2500 $ml$ | 3600 $ml$ | 100 $l$ |

## Solve real life problems involving conversion, addition and subtraction of units of capacity/volume.

**Example 1:** A plastic tank contains 18 $l$ of water. Ali pours 16 $l$ of water into it, how much water will it contain now?

*Solution:*

| | |
|---|---|
| Water in the tank | $\overset{1}{1}8\ l$ |
| Ali pours | $+\ 16\ l$ |
| Total | **34 $l$** |

It will contain 34 $l$ of water in total.

**Example 2:** Two bottles together contain 200 $ml$ of juice. If one of them holds 98 $ml$, how much juice does the other hold?

*Solution:*

| | |
|---|---|
| Two bottles contain | $\overset{1}{2}\overset{9}{0}\overset{10}{0}\ ml$ of juice |
| One of them holds | $-\ 98\ ml$ of juice |
| | **102 $ml$** |

The other bottle holds 102 $ml$ of juice.

## EXERCISE 5.7

**(1) Choose answer in the best unit of volume for the following objects:**

(i)



**4** *l*, **40** *l*, **400** *l*

(ii)



**1** *l*, **1** *ml*, **2** *ml*

(iii)


**500** *l*, **500** *ml*, **800** *ml*

(iv)


**4** *l*, **4** *ml*

(v)


**140** *l*, **140** *ml*

(vi)


**5** *l*, **5** *ml*

2. How much water is left if 19 *ml* are taken from a cup holding 28 *ml*.

3. The bath tub in Sara's house requires 850 *l* of water to fill. It now holds 552 *l*. How many more litres are needed to fill the bath tub?

4. A container contains 98 *l* 300 *ml* of oil. If 51 *l* 700 *ml* more of oil is added to it. How much oil will there be in the tank altogether?

5. A milk van carried 272 *l* of milk. 35 *l* 875 *ml* of the milk were spilt in an accident. How much milk was left?

6. A water drum contains 500 *l* of water. After watering the flowers, 260 *ml* of water is left. How much quantity of water has been used for watering the flowers?

7. There is 1 *l* 10 ml of syrup in a bottle and in another bottle contains 2 *l* 75 *ml*. Find the total quantity of syrup in both bottles.

8. There is 80 *l* 750 *ml* of diesel in the tank of a bus. How much diesel must be added to make it 100 *l*?

## 5.4 TIME

An analogue clock has three hands.
The shorter hand is called **hour hand.**
The longer hand is called **minute hand**.
The thinnest hand is called **second hand**.
The hour hand goes round the clock twice a day.
The minute hand goes round the clock 24 times a day.
The second hand takes 60 little jerks to go round the clock in a minute. We already know that:

| MAY 2014 | | | | | | |
|---|---|---|---|---|---|---|
| Mon | Tue | Wed | Thu | Fri | Sat | Sun |
| * | * | * | 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 19 | 20 | 21 | 22 | 23 | 24 | 25 |
| 26 | 27 | 28 | 29 | 30 | 31 | * |

### TIME MEASUREMENT

| | | | | |
|---|---|---|---|---|
| 1 | hour | = | 60 | minutes |
| 1 | day | = | 24 | hours |
| 1 | week | = | 7 | days |
| 52 | weeks | = | 1 | year |
| 1 | year | = | 12 | months |

### Conversion of units of time

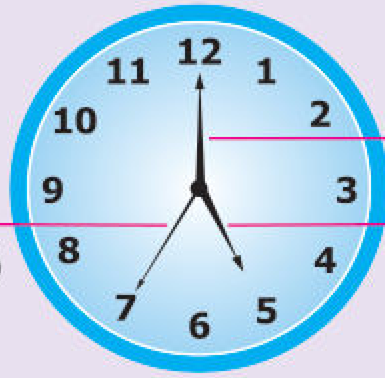
### Read time in hours, minutes and seconds

We have learnt each day has 24 hours .
A day ends at 12 midnight and a new day begins at same time after 12 midnight .

- The time between 12 midnight and 12 noon is called a.m. It means in the late night and morning.
- The time between 12 noon and 12 midnight is called p.m. means in the afternoon, evening and night.

**Teacher's Note** Teacher should revise different time measurements to the students.

Look at this clock. It has 3 hands.




The third hand is long and thin. It moves faster than other two hands. It moves in short jerks. Each jerk made by this hand marks the passing of one second.

The second is the smallest unit of time.

1 minute contains 60 seconds.

**1 minute = 60 seconds**

The 3rd hand takes 60 little jerks to go clock wise round the clock, when 60 seconds are complete a minute has passed.

**How many jerks will the second hand make in one hour?**

**1 hour = 60 minutes**
**or 1 hour = 60 x 60 seconds**
**or 1 hour = 3600 seconds**

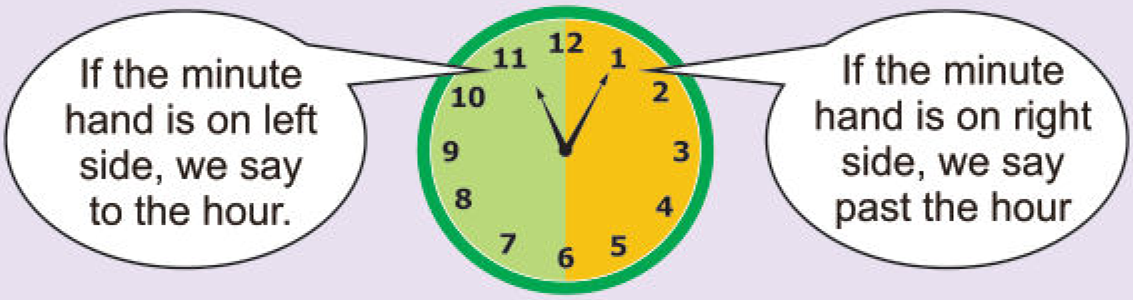
**Activity 1** **Use a.m. or p.m.**

1. 8'o clock in the morning 8 a.m
2. 5'o clock in the evening ______
3. 1'o clock in the morning ______
4. 9'o clock at night ______
5. 2 hours before midnight ______

a.m stands for ante meridien p.m stands for post meridien

# Unit 5 MEASUREMENTS (Time)

**Look at the dial of the clock. We divide it in two parts.**




quarter to 12

11: 45

quarter past 12

12 :15

5 minutes to 4

3 : 55

10 minutes past 6

6 : 10

**Activity 2** **Write time in words and in numbers.**

(i)

5 minutes to 3

2:55

(ii)

10 minutes past 8

8:10

(iii)

(iv)

(v)

(vi)

## Convert hours to minutes and minutes to seconds.

### (a) Convert hours into minutes:

We multiply the number of hours by 60 to convert hours into minutes.

**Example 1:** **Convert 3 hours 15 minutes into minutes**

*Solution:* 3 h 15 m = 3 x 60 + 15
= 180 + 15 = 195 min.

**Activity 1** **Convert the following into minutes.**

(1) **2 h 10 min** = 2 x 60 + 10 = 130 minutes.
(2) **3 h 32 min** = ________ = ________ minutes.
(3) **1 h 45 min** = ________ = ________ minutes.
(4) **5 hours** = ________ = ________ minutes.

### (b) Conversion of minutes to seconds.

We multiply the minutes by 60 to convert into seconds.

**Example 1:** **Convert 4 minutes into seconds.**

*Solution:* 4 min = 4 x 60 sec = 240 seconds

**Example 2:** **Convert 3 minutes 20 seconds**

*Solution:* 3 minutes 20 seconds = 3 x 60 sec. + 20 sec.
= 180 + 20 sec = 200 seconds

**Activity 2** **Convert the following into seconds.**

(1) **47 min** = 47 x 60 = 2820 seconds.
(2) **1 min 5 sec** = ________ = ________ seconds.
(3) **12 min 15 sec** = ________ = ________ seconds.
(4) **45 min 10 sec** = ________ = ________ seconds.
(5) **59 min** = ________ = ________ seconds.

**Teacher's Note** Teacher should explain the students to convert the hour into minutes and minutes into seconds through multiplication by 60.

## Convert years to months, months to days and weeks to days

### (a) Conversion of years to months:

We multiply the number of years by 12 to convert it into months.

**Example 1: Convert 3 years to months**

*Solution:* There are twelve months in a year.

*So,* 3 years = 3 x 12 months = 36 months

**Example 2: Convert 4 years 8 months to months**

*Solution:* 4 years 8 months = 4 x 12 months + 8 months
= 48 months + 8 months = 56 months

**Activity** Convert the following into months.

(1) **2 years** = 2 x 12 = 24 months.
(2) **5 years 2 months** = ______ = ______ months.
(3) **10 years 8 months** = ______ = ______ months.
(4) **7 years 6 months** = ______ = ______ months.

### (b) Conversion of months to days.

We multiply the months by 30 to convert them into days.

**Example 1: Convert 5 months to days**

*Solution:* 5 months = 5 x 30 days = 150 days

**Example 2: Convert 2 months 20 days into days.**

*Solution:* 2 months 20 days = 2 x 30 days + 20
= 60 + 20 days = 80 days

**Activity** Convert the following into days.

(1) **4 months 10 days** = 4 x 30 + 10 = 130 days.
(2) **6 months 4 days** = ______ = ______ days.
(3) **9 months 20 days** = ______ = ______ days.
(4) **18 months 15 days** = ______ = ______ days.